

CSD 211 Computer organization & Architecture

Assignment

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Deadline - 5th November 2024, 11:59PM

Game Description

2048 is a single-player sliding block puzzle game where the objective is to slide numbered tiles on a grid to combine them and create a tile with the number 2048.

Game State Representation

The game state is represented by a 4x4 grid of numbers. Empty cells are denoted by 0. Here's an example state:

	2		4		8		16	
	0		2		4		8	
	2		4		0		4	
	4		2		2		0	

2048 Game Algorithm Explanation

The 2048 game involves sliding all tiles in a chosen direction (up, down, left, or right). Tiles with the same number merge into one when they collide, and merging starts from the last tile in the direction the current move is in; ie, Tiles <2,2,2> sliding to the left will result in tiles <4,2>. Here's how each move works:

Left Move

Before:	After:
2 2 4 4	4 8 0 0
2 2 2 2	4 4 0 0
0 0 2 2	4 0 0 0
2 0 0 2	4 0 0 0

Right Move

Before:	After:
2 2 4 4	0 0 4 8
2 2 2 2	0 0 4 4
0 0 2 2	0 0 0 4
2 0 0 2	0 0 0 4

Up Move

Before:				After:			
2	2	4	4		4	4	4
2	2	2	2		2	0	4
0	0	2	2		0	0	0
2	0	0	2		0	0	0

Down Move

Before:				After:			
2	2	4	4		0	0	0
2	2	2	2		0	0	4
0	0	2	2		2	0	4
2	0	0	2		4	4	4

Lab Goals

In this lab, you will implement a part of the 2048 game in MIPS assembly. Your tasks are:

1. Read the initial game state from a provided text file into memory.
2. Implement the 2048 game algorithm for a specific direction based on your roll number:
 - If (roll_number % 4) == 0: Left Move
 - If (roll_number % 4) == 1: Right Move
 - If (roll_number % 4) == 2: Up Move
 - If (roll_number % 4) == 3: Down Move
3. Apply the implemented move to the game state in memory.
4. Write the resulting game state back to memory.

Your implementation should correctly handle: - Sliding tiles in the specified direction - Merging tiles with the same number - Updating the game state after the move

Remember to comment your code thoroughly.